



MICROTENDER

AI-Augmented Civic Issue and Micro-Tender Management System

A full-stack platform that converts citizen complaints into trackable micro-tenders, connects local vendors, supports workflow monitoring, and demonstrates practical relational database design.

MYSQL

REACT

EXPRESS

Team Members

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Academic Year 2025-26

Project Type

DBMS Mini Project
Web Application + MySQL

1. Abstract

MicroTender is a smart civic-maintenance platform where citizens report public issues, administrators review them, and vendors bid to resolve them. The system unifies complaint registration, tender generation, assignment, work tracking, notifications, anti-fraud logging, and citizen ratings in one structured DBMS application.

2. Problem Statement

- Manual civic complaint handling is slow and hard to monitor.
- Transparency between citizen, authority, and vendor is limited.
- Bidding, work verification, and reporting are often fragmented.
- A relational DBMS workflow is needed for traceability and analytics.

3. Objectives

- Design a normalized database for complaints, tenders, vendors, bids, and ratings.
- Automate complaint-to-tender conversion and lifecycle tracking.
- Support real-time updates, notifications, and completion review.
- Provide secure role-based access for citizen, vendor, and admin users.

4. System Architecture

Frontend Layer

React + Vite
Citizen, Vendor, Admin UI

Application Layer

Node.js + Express
REST APIs + Socket.IO

Data Layer

MySQL Database
Tables, keys, indexes

5. Key Features

Citizen Complaint Filing

Complaint form with category, location, and evidence image.

Micro-Tender Workflow

Complaint-to-review-to-bidding lifecycle with tracked statuses.

Vendor Bidding

Nearby vendors submit bids and estimated completion days.

Notifications and Ratings

Alerts, work updates, completion review, and citizen feedback.

6. Workflow Lifecycle

Citizen submits complaint

System stores complaint and provisional tender

Admin reviews and approves

Nearby vendors bid

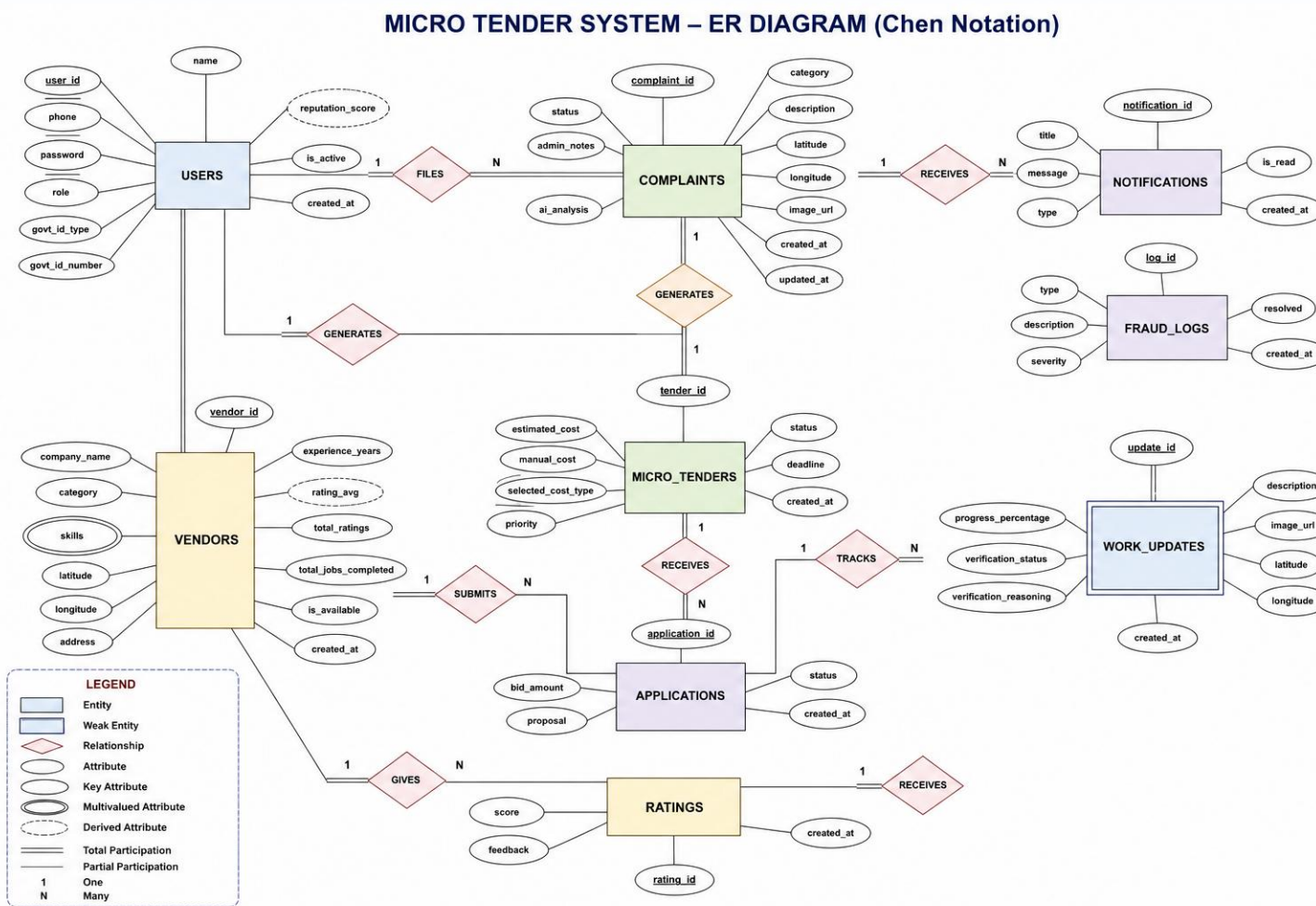
Admin assigns vendor

Vendor posts work updates

Admin verifies completion

Citizen rates resolved work

8. ER Diagram and Database Design



Core Entities

Users, Complaints, Micro_Tenders, Vendors, Applications, Work_Updates, Ratings, Notifications, and Fraud_Logs form the core relational design.

Relationship Logic

A user files complaints, complaints generate tenders, vendors submit applications, tenders receive work updates, and verified work later receives ratings and notifications.

Normalization

- 1NF: Atomic attributes for all core tables
- 2NF: Non-key data depends on the full primary key
- 3NF: Repeated facts are separated into related tables

9. Normalization Summary

- 1NF: Atomic user, complaint, bid, and rating fields
- 2NF: Full functional dependency preserved
- 3NF: Related facts moved into linked tables
- Schema is suitable for reporting and workflow tracking

10. Technology Stack

React 18 Vite 6 Tailwind CSS Node.js Express Socket.IO MySQL JWT Auth
Multer Uploads Gemini + Rule Logic

11. Security Features

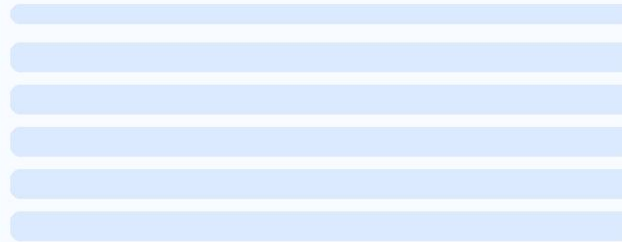
- JWT-based login and role authorization
- Password hashing with bcrypt
- Input validation and upload restrictions
- Fraud logging and audit-friendly workflow tables

12. Interface Snapshots

Citizen Dashboard



Vendor Panel



Admin Analytics



14. Testing and Validation

15 / 15

runtime smoke checks passed

MySQL

database connection verified

Build OK

frontend production build passed

3 Roles

citizen, vendor, admin flows tested

Verified flows include login, complaint submission, approval, vendor bidding, assignment, work updates, admin verification feed, notification read flow, and citizen complaint detail tracking. Note: AI routes fall back when Gemini credentials are invalid, but the DBMS workflow remains functional.

15. Future Scope

Native mobile app for citizens and field vendors

Live GIS mapping and zone-wise analytics dashboard

True AI image verification with valid API integration

Payment gateway and municipal ERP integration